PREFACE

Audio and video content volume has increased dramatically in the digital era, making it extremely difficult for individuals and organizations to manage and understand. Traditional manual methods of summarizing such content are time-consuming and inefficient. Due to this expanding issue, the need for automated solutions that can quickly and effectively transcribe and summarise multimedia information has grown.

This research explores and implements AI-driven audio and video content summarization using STT technology, specifically focusing on leveraging the Whisper model for transcription and advanced summarization techniques. The project aims to create a system that can quickly and effectively summarise audio and video inputs, saving users much time and effort.

The journey of this project has been both challenging and enlightening. It has required an in-depth understanding of NLP and AI and the practical application of these technologies in real-world scenarios. From the initial stages of defining the problem and exploring existing solutions to the meticulous process of implementing and refining the chosen methodologies, this project has provided a comprehensive insight into the capabilities and limitations of current AI technologies.

I thank my supervisors, colleagues, and the academic community for their invaluable support and guidance throughout this project. Their insights and feedback have been instrumental in shaping the direction and outcomes of this work.

This report is structured to provide a detailed account of the project's background, methodology, implementation, and results. This work will contribute to the ongoing advancements in AI and NLP and be a helpful resource for others embarking on similar endeavours. I dedicate this report to all who have contributed to the advancement of artificial intelligence and its applications.